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<Window x:Class="WpfApp1.MainWindow"
        xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
        xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
        xmlns:d="http://schemas.microsoft.com/expression/blend/2008"
        xmlns:mc="http://schemas.openxmlformats.org/markup-
compatibility/2006"
        xmlns:local="clr-namespace:WpfApp1"
        mc:Ignorable="d"
        Title="MainWindow" Height="900" Width="1200"
        WindowStartupLocation="CenterScreen"
        ResizeMode="NoResize">

    <DockPanel>
        <Menu DockPanel.Dock="Top">
            <MenuItem Header="File">
                <MenuItem Header="New"/>
                <MenuItem Header="Open"/>
                <MenuItem Header="Save"/>
            </MenuItem>
            <MenuItem Header="Exit" Click="MenuItem_Exit"/>
        </Menu>

        <Grid>
            <Grid.ColumnDefinitions>
                <ColumnDefinition Width="4*"/>
                <ColumnDefinition Width="1*"/>
            </Grid.ColumnDefinitions>

            <Grid.RowDefinitions>
                <RowDefinition Height="17*"/>
                <RowDefinition Height="3*"/>
            </Grid.RowDefinitions>

            <GroupBox Header="Fourier Plot" Grid.Row="0" Grid.Column="0"
Margin="5">
                <Image x:Name="PlotImage" Stretch="Uniform"/>
            </GroupBox>

            <GroupBox Header="Data Collection" Grid.Row="0"
Grid.Column="1" Margin="5">
                <DataGrid x:Name="MainDataGrid" ItemsSource="{Binding
MyData}" CellEditEnding="MainDataGrid_CellEditEnding"
AutoGenerateColumns="True"/>
            </GroupBox>

            <GroupBox Header="Progress" Grid.Row="1" Grid.Column="0"
Margin="5">
                <ProgressBar x:Name="MainProgressBar" Minimum="0"
Maximum="100" Height="30" Margin="10"/>
            </GroupBox>

            <GroupBox Header="Controls" Grid.Row="1" Grid.Column="1"
Margin="5">

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        <StackPanel VerticalAlignment="Center">
            <Button x:Name="BtnStart" Content="Start" Margin="5"
Click="BtnStart_Click"/>
            <Button x:Name="BtnPause" Content="Pause" Margin="5"
Click="BtnPause_Click"/>
            <Button x:Name="BtnReset" Content="Reset" Margin="5"
Click="BtnReset_Click"/>
        </StackPanel>
    </GroupBox>
</Grid>

</DockPanel>

</Window>

```

```

using System.Collections.ObjectModel;
using System.Text;
using System.Windows;
using System.Windows.Controls;
using System.Windows.Data;
using System.Windows.Documents;
using System.Windows.Input;
using System.Windows.Media;
using System.Windows.Media.Imaging;
using System.Windows.Navigation;
using System.Windows.Shapes;
using System.Windows.Threading; // Konieczne dla DispatcherTimer
using System.Diagnostics;
using System.Security.Policy;

```

```

namespace WpfApp1
{
    /// <summary>
    /// Interaction logic for MainWindow.xaml
    /// </summary>

    public class ForierData
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public double Radius { get; set; }
    }

    public partial class MainWindow : Window
    {
        private void Timer_Tick(object sender, EventArgs e)
        {
            double elapsed = _stopwatch.Elapsed.TotalSeconds;
            double progress = elapsed*100 / TotalTimeSeconds;

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        if (progress >= 100)
        {
            MainProgressBar.Value = 100;
            progress = 100;
            _timer.Stop();

            //rysowanie koła
            radius = MyData[0].Radius;
            DrawCircle(radius);
        }
        else
        {
            MainProgressBar.Value = progress;
        }
    }
}

public ObservableCollection<ForierData> MyData { get; set; }
public DispatcherTimer _timer { get; set; }
public Stopwatch _stopwatch { get; set; }
public const double TotalTimeSeconds = 10.0;
public double radius { get; set; }
public MainWindow()
{
    InitializeComponent();

    _timer = new DispatcherTimer();
    _timer.Interval = TimeSpan.FromMilliseconds(50); //
Aktualizacja co 100 ms
    _timer.Tick += Timer_Tick;
    _stopwatch = new Stopwatch();

    MyData = new ObservableCollection<ForierData>
    {
        new ForierData { Id = 1, Name = "Circle", Radius = 5.0 },
        new ForierData { Id = 2, Name = "Ellipse", Radius = 3.0
    },
        new ForierData { Id = 3, Name = "Spiral", Radius = 7.0 }
    };

    this.DataContext = this;
}

private void MenuItem_Exit(object sender, RoutedEventArgs e)
{
    Application.Current.Shutdown();
}

private void BtnStart_Click(object sender, RoutedEventArgs e)
{
    _timer.Start();
    _stopwatch.Start();
}

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private void BtnPause_Click(object sender, RoutedEventArgs e)
{
    _timer.Stop();
    _stopwatch.Stop();
}

private void BtnReset_Click(object sender, RoutedEventArgs e)
{
    _timer.Stop();
    _stopwatch.Reset();
    MainProgressBar.Value = 0;
    PlotImage.Source = null; // Resetowanie obrazu
    radius = 0;
}

private void DrawCircle(double radius)
{
    int width = 400;
    int height = 400;
    DrawingVisual drawingVisual = new DrawingVisual();
    using(DrawingContext dc = drawingVisual.RenderOpen())
    {
        Point center = new Point(width / 2, height / 2);
        dc.DrawEllipse(null, new Pen(Brushes.Blue, 2), center,
radius, radius);
    }
    RenderTargetBitmap rtb = new RenderTargetBitmap(width,
height, 96, 96, PixelFormats.Pbgra32);
    rtb.Render(drawingVisual);
    PlotImage.Source = rtb;
}

private void MainDataGrid_CellEditEnding(object sender,
DataGridCellEditEndingEventArgs e)
{
    Dispatcher.BeginInvoke(() =>
    {
        if (MyData.Count > 0 && MainProgressBar.Value >= 100)
        {
            radius = MyData[0].Radius;
            DrawCircle(radius);
        }
    }, DispatcherPriority.Background);
}
}

```